Yujiang Li

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Education

Nanjing University | *Computer Science and Technology / B.Eng. (Expected)* 2023.09—2027.06

GPA: 4.23/5.0. Third Prize (Provincial), China Undergraduate Mathematical Contest in Modeling (2025). Best Paper Award, ICSC 2025.

Technical Skills

- **Languages**: Proficient in C/C++, Python, and JavaScript/TypeScript; familiar with Rust and Java; working knowledge of Lua.
- **AI**: PyTorch, MindSpore, and OpenCV; Transformers, multi-agent simulation, and computer vision (SIFT/SSIM); quantization (W8X8/W8), ONNX/ATC deployment, and Ascend NPUs (AscendC/CANN).
- **Systems & Networks**: x86/RISC-V and NEMU; databases (B+ trees, Volcano model); Android privacy auditing (Frida/Appium); protocol stacks, RDMA load balancing, and NS-3; working knowledge of P4_16 / Tofino.
- **Tools**: Linux, Shell, Git, Docker, Wireshark, SQLite, VS Code / (Neo)Vim.

Internship

Intel Connectivity Academy | P4 Teaching Assistant | *Intel Barefoot, Session 19* 2024.03

- Assisted lectures and labs at the Future Network Academy, and completed hands-on P4_16 data-plane programming on the Tofino simulator, covering the programmable switch pipeline and the P4 Studio toolchain.
- Tutored trainees on environment setup and debugging, covering data-plane/control-plane coordination and programmable data-center networks.

Projects

AutoPaper | Local Research Literature Base | *Python, Typer, httpx, SQLite*

- Built a multi-source literature pipeline that unifies records from arXiv, Crossref, OpenAlex, PubMed, and Semantic Scholar, deduplicates by DOI and normalized title into SQLite, and continues when any single source fails.
- Keywords are matched literally, with any/all modes and loose or strict field filters. With no model configured, results rank by coverage, title, citations, and year; with embeddings, synonymous papers are recalled and only the top abstracts are rescored. Hits below the threshold are marked deferred.
- Archives local PDF/TXT/Markdown and recovers metadata from the first page. Runs auditable close reading against a fixed question set and flags claims the paper does not state. Draws an in-browser knowledge graph by note relevance, exports shortlists and topic reports, and is covered by pytest.

Gift Flow | Multi-Agent Sociological Simulation | *TypeScript, Node.js, React, PixiJS, DeepSeek*

- Inspired by Yan Yunxiang's *The Flow of Gifts*, turned delayed reciprocity, vertical asymmetry, and prestige attraction into falsifiable hypotheses, and built a full village simulation: event trigger, dialogue, rule settlement, log snapshot, and pixel replay (GitHub).
- Designed a hybrid architecture in which the LLM only proposes and a RuleEngine is the sole writer of world state (cash, relationships, social face, and gift ledgers). DeepSeek emits utterances and intents, which keeps each run consistent and reproducible.
- Modeled agents with BDIE, a cognitive-tree memory, and Big Five–derived traits. Main run: 100 agents × 365 days (reciprocity 0.90, default rate 0.022, gift–prestige correlation 0.51), with offline pixel step-replay in React and PixiJS.

Dajian | Chinese Calligraphy Style Recognition | *Python, PyTorch, OpenCV, SQLite*

- Designed a style-analysis pipeline: OCR, per-character retrieval, calligrapher attribution, and an eight-dimension style report, exported in one step to JSON, HTML, and a radar chart.
- Combined SIFT, template matching, and SSIM for glyph similarity, with Hu-moment prefiltering and a SQLite feature cache. Deduplicated characters, then attributed authors by confidence-weighted voting and alias normalization.

- Trained a handwritten Chinese recognizer on CASIA-HWDB with PyTorch and connected it as a local OCR backend, alongside Baidu's cloud API. An optional LLM can draft a Chinese style commentary.

Android Privacy Audit Agent | Evidence-Driven Audit |*Python, Frida, Appium, Androguard*

- Designed an evidence-first audit pipeline: privacy-policy parsing, static reversing, dynamic forensics, multi-source fusion, and rule-based decisions. Reports in JSON, HTML, and Excel cover the rights to be informed, to consent, and of access.
- Static track: permissions, trackers/SDKs, and sensitive APIs via Androguard and Exodus. Dynamic track: Frida hooks, Appium, and mitmproxy, with fallback when Frida fails.
- A YAML rule engine issues the final verdict, and DeepSeek assists UI-path navigation. Batch-audited 9+ apps, including Keep, Xiaohongshu, and Alipay; the right-to-be-informed audit succeeded on 9/9.

Social Network Condensation | Research |*Best Paper Award, ICSC 2025* 2024.10—2025.01

- Contributed to *Preserving Topological Information for Social Network Condensation via Knowledge Distillation*, which received the **Best Paper Award** at ICSC 2025 (6th National Conference on Social Networks).
- Responsible for GNN-to-MLP knowledge-distillation optimization. Trained, ran, and tuned models with PyTorch and Transformers, and studied graph condensation and topology-preserving methods.

DoJot | Mahjong Soul Web-Client AI |*JavaScript, Python, Bayesian optimization*

- Built a real-time match AI as a Tampermonkey userscript: it parses table state and decides discards, melds, riichi, and folds. A strategy state machine routes standard hands, seven pairs, thirteen orphans, and a fold line.
- Scored discards as approximate two-turn efficiency plus expected value, minus a danger penalty, and switched between attack and defense with suji and wall-tile heuristics. Iterated weights with GPR and expected improvement. In about four days of Gold Room four-player play, rank rose from Expert toward Master.

NEMU / NJUDB | Systems and Database Coursework |*C / C++* 2024.03—2024.12

- **NEMU**: Independently implemented a RV32 teaching-subset emulator: decode and execute, memory and device abstraction, system calls, time-sharing, and virtual memory. Ran *Chinese Paladin* on it.
- **NJUDB**: Implemented an LRU-K buffer manager, a Volcano-model query engine, and a full B+ tree, with equality and range scans, nested-loop joins, aggregation, and external merge sort.

Intelligent Computing Systems | Ascend |*NumPy, MindSpore, CANN* 2025.03—2025.06

- Implemented a 3-layer fully connected net and a full DistilBERT in NumPy (91.08% on AG News). Quantized TinyLlama on the Ascend 310B and ran on-device inference via ONNX/ATC, with KV cache and top-k sampling.
- Pretrained a GPT with MindSpore on Ascend 910 and added stylized continuation. Implemented L1Loss, Softplus, and Matmul in AscendC, including host/kernel shape inference, tiling, and double-buffered pipelines.

Data-Center Load Balancing | Coursework |*NS-3, Docker, Python* 2025.09—2025.12

- Read RDMA load-balancing papers and the limits of ECMP, flowlet switching, and packet spraying for continuous flows on leaf-spine and fat-tree fabrics.
- Extended NS-3 ConWeave with reduce-broadcast All-Reduce traffic, and automatically compared FECMP, CONGA, LetFlow, and ConWeave on throughput imbalance and FCT slowdown.

Summary

- Coursework and projects across systems, AI, databases, and networking. Provincial top scorer in Gaokao English; reads English technical documents and papers fluently.
- Interested in LLM optimization, multi-agent systems, and embodied intelligence. Led and collaborated on substantial projects, and can deliver independently or with a team.
- **IELTS**: Listening 6.0, Reading 7.0, Writing 6.0, Speaking 5.5 (overall 6.0).